Answer **all** the questions in this Section in the spaces provided.

1 A vehicle travels from one traffic junction to another. Its acceleration-time (*a*-*t*) graph is shown in Fig. 1.1.

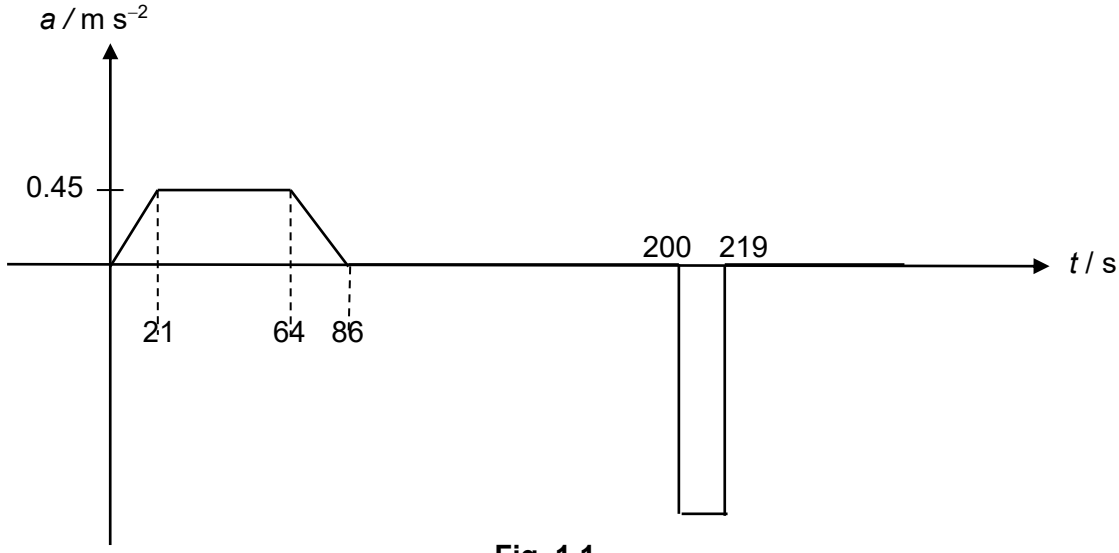

Fig. 1.1

(a) Use Fig. 1.1 to determine the average acceleration of the vehicle over the first 200 s.

average acceleration = ... m s^{-2} [2]

(b) Given that the velocity of the vehicle is 0 m s^{-1} at t = 0 s and t = 219 s, determine

(i) the magnitude of its maximum velocity, and

magnitude of maximum velocity = ... m s^{-1} [2]

(ii) sketch its corresponding velocity-time (v-t) graph from t = 0 s to t = 219 s in Fig 1.2.

[4]

Fig. 1.2